variables proofs, delicately prodding at their component pieces until he found one that easily snapped in half—an unjustified assumption at the foundation. Turning this assumption on its head, Bell showed that the purported "no-hidden-variables" proofs collectively suggested something else entirely, something the original creators of these proofs did not intend or fully understand. Specifically, Bell found that a hidden-variables theory could avoid the traps laid by these proofs if it had a rather peculiar property, later dubbed *contextuality.*

Contextuality means that the outcome of a measurement on a quantum system depends on the other things you measure about that system at the same time. In other words, if you measure a property of a thing, the outcome of your measurement can depend on what other stuff you measure about that thing at the same time. In a contextual world, if you measure the energy of a neutron along with its momentum, you'll get an answer about the neutron's energy—but if you had measured the energy along with the location, you could have gotten a completely different answer about the neutron's energy, simply by virtue of the context in which you made the energy measurement.

To get a better handle on contextuality, let's forget about neutrons and instead talk about something larger and more familiar: a roulette wheel. Imagine your friend Flo is at a roulette wheel in a casino, and you're talking with her over the phone. You can't see the wheel, but you can ask her questions about how the ball landed in the wheel after any particular spin. You can ask whether the ball landed on an even or odd number, a high or low number, or a red or black number. (Roulette wheels are constructed in such a way that half the numbers are red and half are black, but they're not split along evens-odds or highs-lows—half the highs and half the lows are red, and the same is true for evens and odds. See Figure 7.2.) But Flo is being strangely reticent to tell you what's happening at the casino: for any spin of the wheel, she'll only tell you the answers to two of your questions, not all three. Normally, you wouldn't think this mattered: no matter what Flo tells you, the ball landed in a particular slot on each spin. Therefore, even though the ball's actual state is hidden from you, the answers to all three of your questions are already set once the ball stops moving. If the ball lands on 34, then the ball landed on a high, even, red number, even if Flo will tell you only two of those three things.